Diffractive Practices to Slow Time

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Concerned with developing new ways of reading, writing, and thinking with high school students about texts, the authors of this article—a creative writing teacher and an experimental art teacher, who are also doctoral students—seek to use diffraction, as developed and theorized by Donna Haraway and Karen Barad specifically, to slow down the frenetic pace of the classroom through the entanglement of the radical and subversive possibilities of blackout poetry and computational art represented by coding. The article explores this entanglement of blackout poetry and poetic coding through the interference created and experienced by students and teachers in a classroom as they use the tangible matter of texts to see and re-see what mattered *in those texts through the randomness and agency afforded by markers and code. The article suggests that such activities slow down classrooms, allowing students to take time with texts, offering them opportunities to re-imagine and critique existing texts through an engaging classroom activity that blends old and new processes.*

KEYWORDS: Diffraction, creative writing, art education

The classroom purrs. Paper ripping, markers squeaking, computer keys tap tap, and students murmur to create a rich timbre of layered sound. Amy sits on the floor and writes lines of computer code. Students look up from their blackout poems to watch the code and its progeny cascade down and crowd the large classroom screen. Stories emerge from the entanglement. Nick moves around the room, replenishing exhausted markers, commenting on student work, and pausing frequently to notice what appears and disappears on the screen.

Background

The two authors, both teachers and PhD students, deconstruct texts using randomness and acts of diffraction to discover, with their students, new methods of reading, writing, and thinking. Rapid technological intensification, ever-evolving testing requirements, and curricular violence through book banning means that time for "sharing and communal learning . . . for reflection and open-ended enquiry" is rapidly disappearing in K–12 spaces—where it is perhaps most needed (Braidotti, 2019, p. 148). There is no time for Deleuzian novelty or Dewey's joy. In this computation-meets-poetry project, a creative writing teacher and an experimental art and technology teacher came together during a creative writing class in a large urban public school to slow down and diffract together through the nodes of entanglement between the subversive act of blackout poetry and the radical possibilities of computational art.

Time spent with texts should be slow and purposeful, yet vigorous. Text is alive. Reading should be done with the text as an agential partner, not as a corpse to be dissected. A languorous reading of text should be the heart of an English Language Arts classroom. Time to read, share, discuss, debate, question, write about, and to immerse oneself in a text is "a qualitatively different use of time that is not indexed on short-term and fast productivity" (Braidotti, 2019, p. 148). Our project aims to "honour complexity and resonate with a more embodied and embedded vision of intelligence, as a situational and affective quality" (Braidotti, 2019, p. 148).

Our intent with the entangled poeticode presented here was to operationalize Karen Barad's notion of diffraction as "reading insights through one another in attending to and responding to the details and specificities of relations of difference and how they matter" (Barad, 2007, p. 71). We envision systems of entanglement to explore how students can question and trouble existing texts and, in the process, reveal radical or subversive readings that examine "patterns of interference that exist beyond purely anthro-dialectical thought" (Johnston, 2023, p. 7).

Blackout Poetry

Blackout poetry is now a familiar exercise in the English language arts (ELA) classroom. It consists of providing students with existing texts—often newspapers, magazines, and books—and giving them the time and space to destroy, deface, and remake the original texts into something new, most commonly, a poem. Rather than thinking of blackout poetry as extractive, like mining, it is perhaps more accurate to think of it archaeologically, with the student carefully brushing away dirt and time to bring out in relief the bones of some new animal whose existence was hitherto unknown or unacknowledged.

Coding

Computational art and creative coding have existed for over 70 years. Computation is an art medium, and creative coding is the method of artistic expression. Amy writes custom computer programs to make art. Reading text into a computer as data allows the affordances of computational art to realign and re-imagine the text through randomness and generativity, using the computer and code as a semiotics engine (Nake, 2008). Code allows us to choose not to publish or concretize our thoughts, committed to only one linear configuration of text, but to live in the fleeting moments of joy and epiphany as the words float across the screen. We can anchor ourselves in the moment, relishing the voice of each word as it diffracts, sending ripples of interference like raindrops hitting still water.

In creative coding, artists frequently begin their work by utilizing template or example code provided by other artist coders. The original example code used for this project has words falling like rain (Shiffman, 2023). The challenge has been to allow each word to bear weight. Getting the words on the screen in a pleasing fashion is not unlike finding the sweet spot between order and chaos (Nake, 2008). As posthumanist artists and scholars, we ask ourselves questions like: In each configuration of the code, is it rude to discriminate against the most common words like "the"? If there are one hundred instances of "the" in a text, does each instance exist individually or collectively?

A key component of both blackout poetry and computational art is how randomness becomes a tool. Students select the text to be blacked out—a book, for example—then choose a page, often at random, to work on. A single page of text will reveal vastly different poems based on who and/or when the blacking out takes place, with each iteration revealing something that did not exist prior to this particular intra-action.

Process: Diffracting in Class

In the case of blackout poetry or poetic coding, the interference with the text acts as a generative tool as a new text is created, one that entangles physical *matter* (the book, the marker, the computer) with the new meaning of the poem acquired through the poet/student's choices—their sense of what *matters* in the text. Scissors and keyboards become apparatchik co-conspirators in the agential cut. The cuts and choices of the student revolt against the authority of the original author, with the new words, phrases, and arrangements able to defy the original author's intentions. Figure 1 demonstrates an example where a seemingly neutral historical text is made to comment on the concepts of authority and human rights. The original text still exists (and is often still visible, ghost-like behind the veil of the

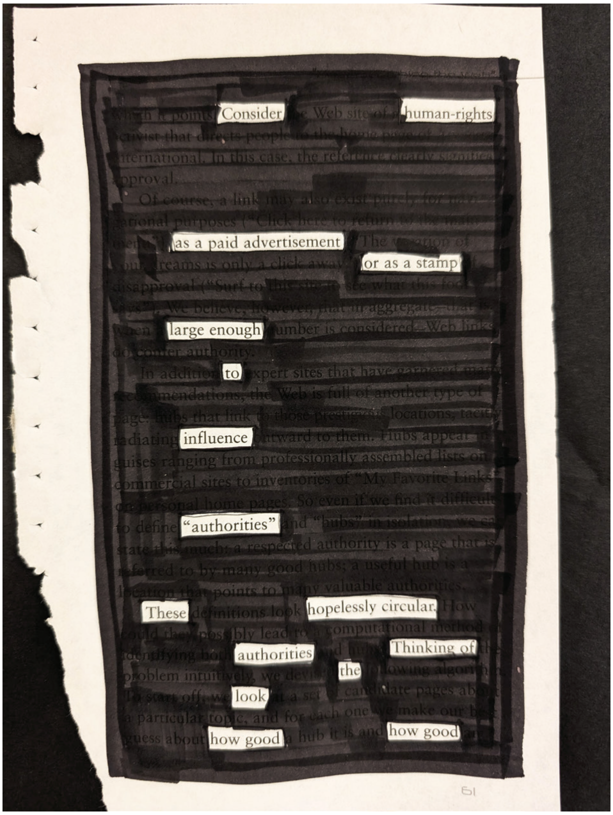

Figure 1. Nick DePascal, *Example of Blackout Poetry*.

marker) at the same time as a new text is created; they exist simultaneously, toying with our notions of either/or, and creating new cyborg offspring (Haraway, 1991) that do not follow the rules of either progenitor.

Similarly, coding runs in real-time, created and re-created at the poet's whims, offering flexibility and opportunity for happy accidents of generativity as the writer re-imagines the text. Coding offers an additional flexibility because with only a few keystrokes, the poeticoder can instantaneously change the parameters of what appears on their screen. Figure 2 shows the results of code that processed four different texts simultaneously: *Dracula*, *Frankenstein*, *The Yellow Wallpaper*, and *The Metamorphosis*. The code searched through each text for the most frequent words used by the author, then compared that list of most frequent words with the lists of frequent words from a paired text. *Frankenstein* was paired with *Dracula*, and T*he Yellow Wallpaper* was paired with *Metamorphosis*. Only the words that are unique to that author are displayed on screen. Blackout poetry and the visualization of coding presented here can entangle even further, as there is opportunity to visually black out words,

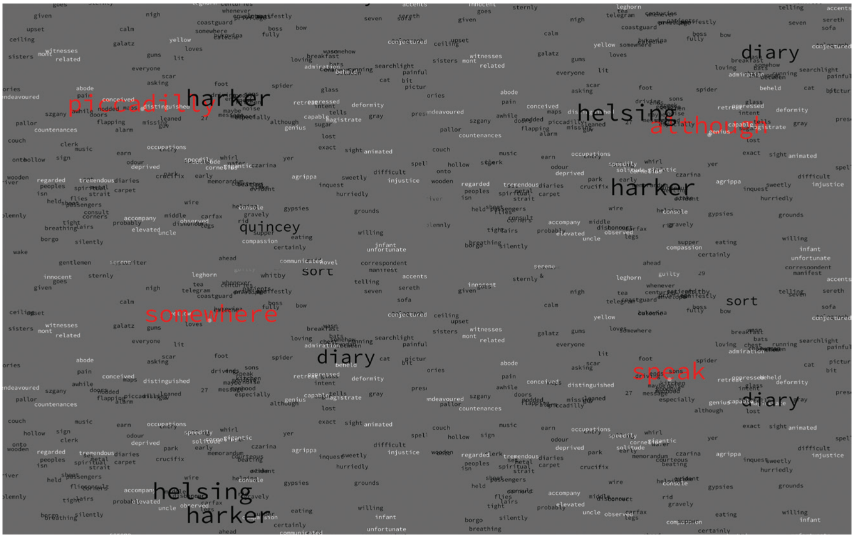

Figure 2. Amy Traylor, *The Results of Code Processed From Four Different Texts.*

either from printed versions of the text, or by doing it live on the screen within the classroom. Such projects offer multiple means of publication, in hard copy, digitally, or students can also refuse to concretize their thoughts at all and let the words stay hidden except for whenever the code runs.

Implications/Provocations

Using blackout poetry and coding diffractively offer students new tools with which to see, re-see, and critique texts—though learning and practicing take *time*. In teaching and in thinking about how to slow down the classroom and the frenetic pace of schooling, we offer these entangled and radical possibilities of using poetry and code as provocative pedagogical methods of critique and wonder. There are many resources available to access creative coding tools. Amy utilized a programming language called Processing (Fry & Reas, 2011), which was developed in 2001 at the MIT Media Lab explicitly for artists and designers to more easily access computer programming for their work. A newer sister project, p5.js, is also common in classrooms that code. Both tools are open source, completely free, and have a multitude of resources available for educators.

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